that they've rushed apart at faster than the speed of light. And, indeed, the colossal outward push of repulsive gravity during the inflationary phase *did* drive every region of space away from every other at much faster than the speed of light. Again, this offers no contradiction with special relativity, since the speed limit set by light refers to motion through space, not motion from the swelling of space itself. So a novel and important feature of inflationary cosmology is that it involves a short period in which there is superluminal expansion of space.

17. Note that the numerical value of the critical density decreases as the universe expands. But the point is that if the actual mass/energy density of the universe is equal to the critical density at one time, it will decrease in exactly the same way and maintain equality with the critical density at all times.

18. The mathematically inclined reader should note that during the inflationary phase, the size of our cosmic horizon stayed fixed while space swelled enormously (as can easily be seen by taking an exponential form for the scale factor in note 10 of Chapter 8). That is the sense in which our observable universe is a tiny speck in a gigantic cosmos, in the inflationary framework.

19. R. Preston, *First Light* (New York: Random House Trade Paperbacks, 1996), p. 118.

20. For an excellent general-level account of dark matter, see L. Krauss, Quintes*sence:The Mystery of Missing Mass in the Universe* (New York: Basic Books, 2000).

21. The expert reader will recognize that I am not distinguishing between the various dark matter problems that emerge on different scales of observation (galactic, cosmic) as the contribution of dark matter to the cosmic mass density is my only concern here.

22. There is actually some controversy as to whether this is the mechanism behind all type Ia supernovae (I thank D. Spergel for pointing this out to me), but the uniformity of these events—which is what we need for the discussion—is on a strong observational footing.